ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 10

PHYSICAL SCIENCES P1

MARKING GUIDELINE

MARKS: 150

This marking guideline consists of 5 pages.

Written by Academic Central for practice, following the CAPS curriculum for Grade 10. It is not a Department of Basic Education marking guideline.

How to mark: each ✓ is one mark. Accept correct alternative methods. Give follow-through marks (CA) where a learner correctly uses an earlier wrong answer. Penalise once per question for missing or wrong units in final answers.

QUESTION 1

1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	1.10
C ✓✓	A ✓✓	B ✓✓	C ✓✓	C ✓✓	D ✓✓	D ✓✓	B ✓✓	B ✓✓	A ✓✓

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QUESTION 2

2.1 A physical quantity that has both magnitude ✓ and direction ✓. (2)

2.2 The change in position of an object ✓✓ (a straight line from start to end, with direction). (2)

2.3 Scalar ✓ (1)

2.4 $300 + 400 = 700\ \text{m}$ ✓ (1)

2.5 $\Delta x^2 = 300^2 + 400^2$ ✓ $\rightarrow \Delta x = \sqrt{250\,000}$ ✓ $= 500\ \text{m}$ ✓ (3)

2.6 $\tan\theta = \frac{400}{300}$ ✓ $\rightarrow \theta = 53{,}13°$ north of east ✓ (2)

2.7 $\text{average speed} = \frac{\text{distance}}{\text{time}} = \frac{700}{500}$ ✓✓ $= 1{,}4\ \text{m·s}^{-1}$ ✓ (3)

2.8 $v_{ave} = \frac{\Delta x}{\Delta t} = \frac{500}{500}$ ✓ $= 1\ \text{m·s}^{-1}$ ✓ at 53,13° north of east ✓ (3)

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QUESTION 3

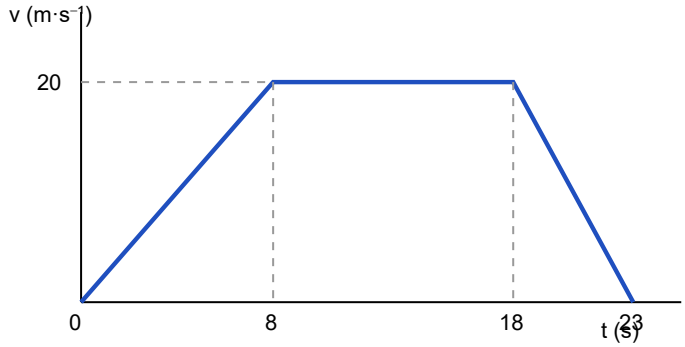
3.1 The rate of change of velocity ✓✓. (2)

3.2 $v_f = v_i + a\Delta t = 0 + (2{,}5)(8)$ ✓✓ $= 20\ \text{m·s}^{-1}$ (in the direction of motion) ✓ (3)

3.3 $\Delta x = v_i\Delta t + \frac{1}{2}a\Delta t^2 = 0 + \frac{1}{2}(2{,}5)(8)^2$ ✓✓ $= 80\ \text{m}$ ✓ (3)

3.4 $a = \frac{v_f - v_i}{\Delta t} = \frac{0 - 20}{5}$ ✓✓ $= -4\ \text{m·s}^{-2}$ ($4\ \text{m·s}^{-2}$ opposite to the motion) ✓ (3)

3.5 (4)



Marks: straight line from (0 ; 0) up to 20 m·s⁻¹ at 8 s ✓; horizontal line from 8 s to 18 s ✓; straight line down to 0 at 23 s ✓; values and labelled axes ✓

3.6 $\text{Displacement} = \text{area under graph}$ ✓ (4)

$= ½(8)(20) + (10)(20) + ½(5)(20)$ ✓ $= 80 + 200 + 50$ ✓ $= 330$ m ✓

3.7 $v_{ave} = \frac{\Delta x}{\Delta t} = \frac{330}{23}$ ✓✓ $= 14{,}35$ m·s^{-1} in the direction of motion ✓ (3)

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QUESTION 4

4.1 In the absence of air resistance (or any external forces), the total mechanical energy of an object remains constant (is conserved) ✓✓. (2)

4.2 $E_P = mgh = (0{,}5)(9{,}8)(20)$ ✓✓ $= 98$ J ✓ (3)

4.3 $E_M = 98$ J at all heights ✓ (4)

E_P at 5 m $= (0{,}5)(9{,}8)(5) = 24{,}5$ J ✓

$E_K = 98 - 24{,}5$ ✓ $= 73{,}5$ J ✓

4.4 $E_K = E_M = 98$ J at the ground ✓ (4)

$½(0{,}5)v^2 = 98$ ✓ $\rightarrow v^2 = 392$ ✓ $\rightarrow v = 19{,}80$ m·s^{-1} ✓

4.5 $E_K = ½mv^2 = ½(1\,200)(20)^2$ ✓✓ $= 240\,000$ J $(2{,}4 \times 10^5$ J) ✓ (3)

4.6 It becomes 4 times bigger ✓✓ ($E_K \propto v^2$, and $2^2 = 4$). (2)

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QUESTION 5

5.1 A wave in which the particles of the medium vibrate perpendicular ✓ to the direction in which the wave moves ✓. (2)

5.2 $v = f\lambda = (50)(0{,}4)$ ✓✓ $= 20$ m·s^{-1} ✓ (3)

5.3 $T = \frac{1}{f} = \frac{1}{50}$ ✓✓ $= 0{,}02$ s ✓ (3)

5.4 When two pulses meet, the resultant displacement is equal to the algebraic sum of the displacements of the individual pulses ✓✓. Amplitude = 3 + 2 = 5 cm (constructive interference) ✓ (3)

5.5 $\text{Distance travelled} = 2 \times 170 = 340 \text{ m} \checkmark$ (3)

$$v = \frac{\text{distance}}{\text{time}} = \frac{340}{1,0} \checkmark = 340 \text{ m·s}^{-1} \checkmark$$

5.6 Longitudinal ✓ (1)

5.7 Frequency ✓ (1)

5.8 No ✓. Humans can only hear frequencies from about 20 Hz to 20 kHz, and 25 kHz is above this range (ultrasound) ✓. (2)

[18]

QUESTION 6

6.1 Radio waves, microwaves, visible light, X-rays ✓✓ (2)

6.2 A packet (quantum) of energy ✓ of electromagnetic radiation ✓. (2)

6.3 $E = hf = (6,63 \times 10^{-34})(6 \times 10^{14}) \checkmark\checkmark = 3,98 \times 10^{-19} \text{ J} \checkmark$ (3)

6.4 $\lambda = \frac{c}{f} = \frac{3,0 \times 10^{8}}{6 \times 10^{14}} \checkmark\checkmark = 5 \times 10^{-7} \text{ m} \checkmark$ (3)

6.5 Any ONE: sunburn, skin cancer, damage to the eyes (cataracts), premature ageing of the skin ✓ (1)

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QUESTION 7

7.1 The net charge of an isolated system remains constant ✓✓ (charge cannot be created or destroyed). (2)

7.2 $Q = \frac{(+8) + (-4)}{2} \checkmark = +2 \text{ nC on each sphere} \checkmark$ (2)

7.3 $\Delta Q \text{ on B} = (+2) - (-4) = 6 \text{ nC} \checkmark$ (4)

$$n = \frac{\Delta Q}{q_e} = \frac{6 \times 10^{-9}}{1,6 \times 10^{-19}} \checkmark = 3,75 \times 10^{10} \text{ electrons} \checkmark$$

Electrons moved from sphere B to sphere A ✓

7.4 The balloon polarises the wall: electrons in the wall are repelled, so the surface near the balloon becomes positive ✓. Opposite charges attract, so the balloon sticks ✓. (2)

7.5 Charge builds up on the truck from friction ✓. Earthing lets the charge flow safely to the ground, so no spark can ignite the fuel ✓. (2)

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QUESTION 8

8.1 The rate of flow of charge ✓✓. (2)

8.2 $\frac{1}{R_p} = \frac{1}{6} + \frac{1}{3} \checkmark = \frac{1}{2} \checkmark \rightarrow R_p = 2 \ \Omega \checkmark$ (3)

8.3 $R_{total} = 4 + 2 = 6 \ \Omega \checkmark$ (1)

8.4 $I = \frac{V}{R} = \frac{12}{6}$ ✓✓ $= 2\ A$ ✓ (3)

8.5 $V = IR = (2)(4)$ ✓✓ $= 8\ V$ ✓ (3)

8.6 $V_{parallel} = 12 - 8 = 4\ V$ ✓; $I_3 = \frac{V}{R} = \frac{4}{3}$ ✓ $= 1{,}33\ A$ ✓ (3)

8.7 $Q = I\Delta t = (2)(2 \times 60)$ ✓✓ $= 240\ C$ ✓ (3)

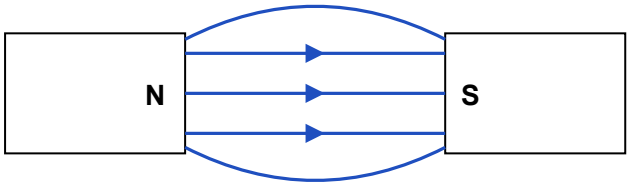
8.8 DECREASE ✓. The total resistance increases (4 + 3 = 7 Ω) ✓, so for the same emf the current is smaller (I = V/R = 12/7 = 1,71 A) ✓. (3)

8.9 In parallel across R_1 ✓ (1)

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QUESTION 9

9.1 (3)



Marks: field lines from N to S between the poles ✓; lines do not cross and are closest where the field is strongest (straight and evenly spaced in the middle) ✓; arrows pointing from N to S ✓

9.2 The Earth's geographic North Pole is close to a magnetic SOUTH pole ✓. Opposite poles attract, so the north pole of the compass needle points there ✓. (2)

9.3 Any ONE: place compasses around the wire / sprinkle iron filings on a card around the wire (Oersted's experiment) ✓ (1)

9.4 Point the thumb of the right hand in the direction of the (conventional) current ✓; the curled fingers show the direction of the magnetic field ✓. (2)

9.5 Any TWO: compass, electric motors, generators, loudspeakers, MRI scanners, fridge magnets, magnetic strips on bank cards, scrapyard electromagnets ✓✓ (2)

[10]

TOTAL: 150

END OF MARKING GUIDELINE